

Lab 2 : MultiCycle Performance

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Contents

0.1	Matrix Multiplication Performance	2
0.1.1	Procedure	2
0.1.2	Results	2
0.2	Testall Performance	5
0.2.1	Procedure	5
0.2.2	Results	5
0.3	Notes	8
0.3.1	RARS	8

0.1 Matrix Multiplication Performance

0.1.1 Procedure

1. For each matrix test case (3x3, 10x10, 16x16, 50x50) ...
2. Compile the matrix multiplication program in the OTTER-memory mapped version of RARS. The RARS program should compile but not run any instructions of assembly. The machine code file needs to be saved as a .mem
3. Load the mem file into the multicycle OTTER in Vivado and simulate.
4. Capture the waveforms for the matrix multiplication to prove the correct execution.
5. Provide CPU execution time , Power, and Area utilization.

0.1.2 Results

All of the RARS matrix multiply compiles start and end with the same instruction address. For timing purposes, the hex address 0x54 for start and 0x1E0 will be used for the stop.

Text Segment					
Blkpt	Address	Code	Basic	Source	
	0x00000040	0x00048593	addi x11,x0,0	737:	addi a1, s1, 0 # a1 is base addr of mat B
	0x00000040	0x00090613	addi x12,x18,0	738:	addi a2, s2, 0 # a2 is base addr of mat C
	0x00000050	0x00300093	addi x13,x0,3	739:	addi a3, zero, 3 # a3 is one dim of the mat
	0x00000054	0x02c000ef	jal x1,0x0006002c	740:	jal matmul # no return val: C is in mem already
	0x00000055	0x01c12503	lw x10,28(x2)	743:	lw a0, 28(sp)

Figure 1: MatrixMult.asm Start Codes for PC

Text Segment					
Blkpt	Address	Code	Basic	Source	
	0x00000148	0xff5ff06f	jal x0,0xffffffff4	922:	while
	0x0000014c	0x00008057	jalr x0,x1,0	926:	ir ra
	0x0000014d	0x0a008939	addi x17,x0,10	931:	li a7, 10 # Load system call number for exit()
	0x000001e4	0x00000513	addi x10,x0,0	932:	li a0, 0 # Load the exit status code (EXIT_SUCCESS)
	0x000001e8	0x00000073	ecall	933:	Make the system call exit() to term prog

Figure 2: MatrixMult.asm Exit Codes for PC

TC 3x3

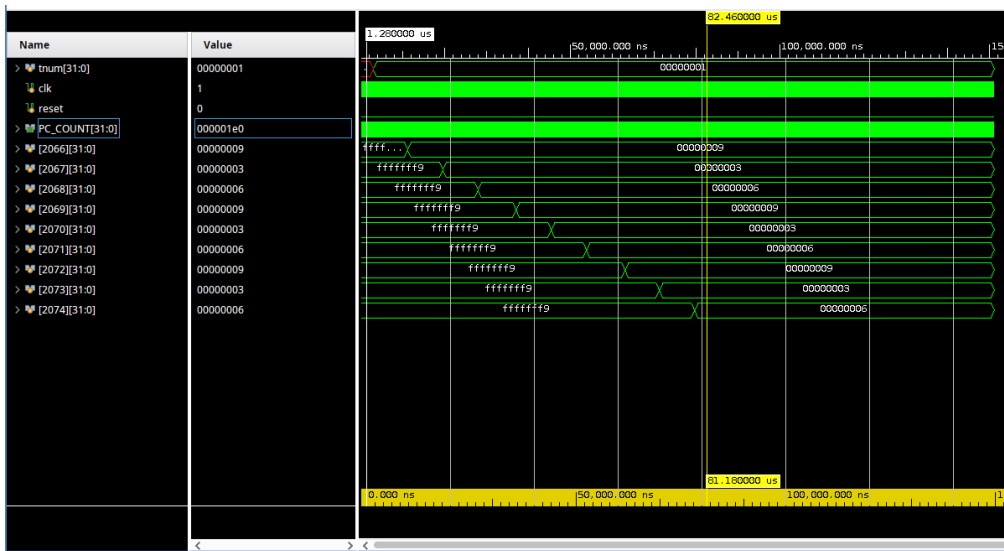


Figure 3: Waveform and Timing of TC 3x3 mat

$$\begin{aligned} t_{execution} &= t_{end} - t_{start} \\ &= 82.460\mu s - 1.280\mu s \\ &= 81.18\mu s \end{aligned}$$

$$\begin{aligned} n_{CLK_Cycles} &= \frac{t_{execution}}{T_{CLK}} \\ &= \frac{81.18\mu s}{10ns} \\ &= 8,118 \end{aligned}$$

TC 10x10

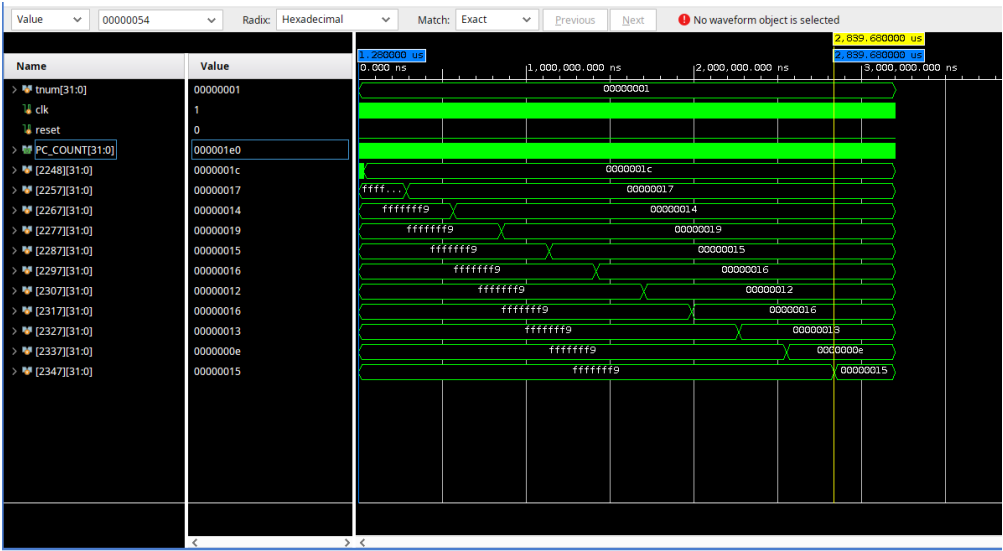


Figure 4: Waveform and Timing of TC 10x10 mat

$$\begin{aligned} t_{execution} &= t_{end} - t_{start} \\ &= 2.839680ms - 1.28\mu s \\ &= 2.8384ms \end{aligned}$$

$$\begin{aligned} n_{CLK_Cycles} &= \frac{t_{execution}}{T_{CLK}} \\ &= \frac{2.8384ms}{10ns} \\ &= 283,840 \end{aligned}$$

TC 16x16

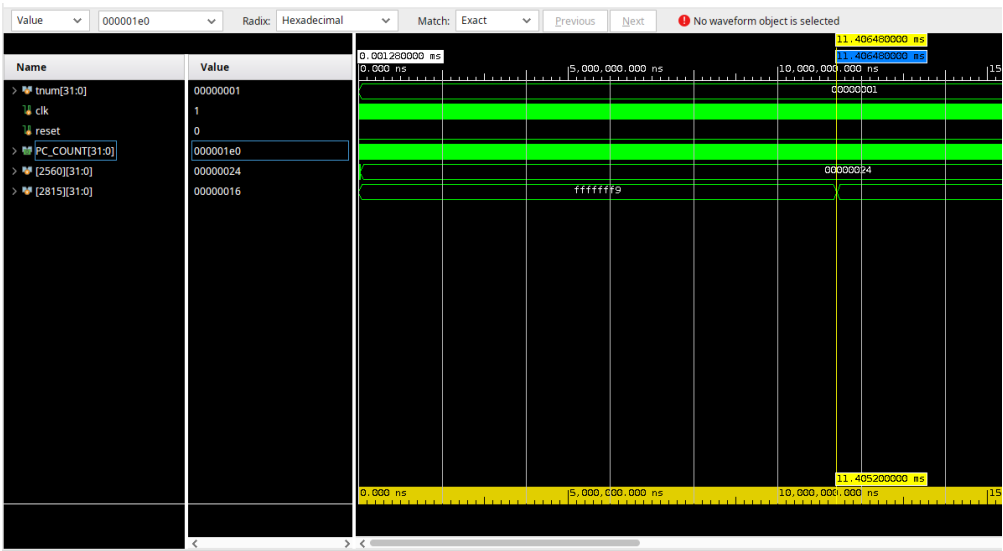


Figure 5: Waveform and Timing of TC 16x16 mat

$$\begin{aligned} t_{execution} &= t_{end} - t_{start} \\ &= 11.40648ms - 0.00128ms \\ &= 11.4052ms \end{aligned}$$

$$\begin{aligned}
 n_{CLK_Cycles} &= \frac{t_{execution}}{T_{CLK}} \\
 &= \frac{11.4052ms}{10ns} \\
 &= 1,140,520
 \end{aligned}$$

MatrixMult Summary

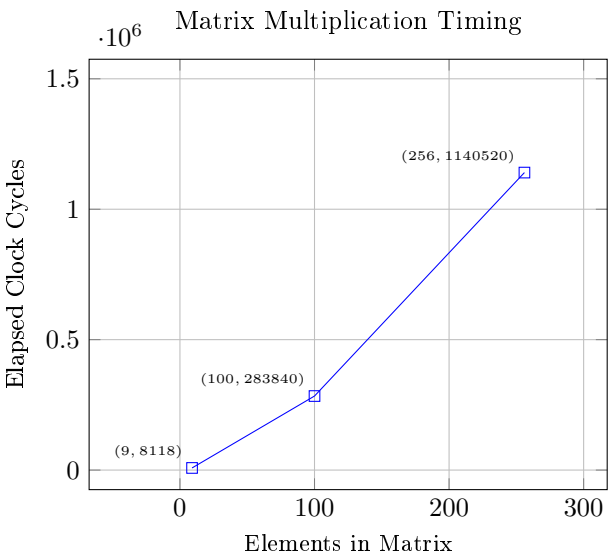


Figure 6: Computation Trend for Matrix Multiplication

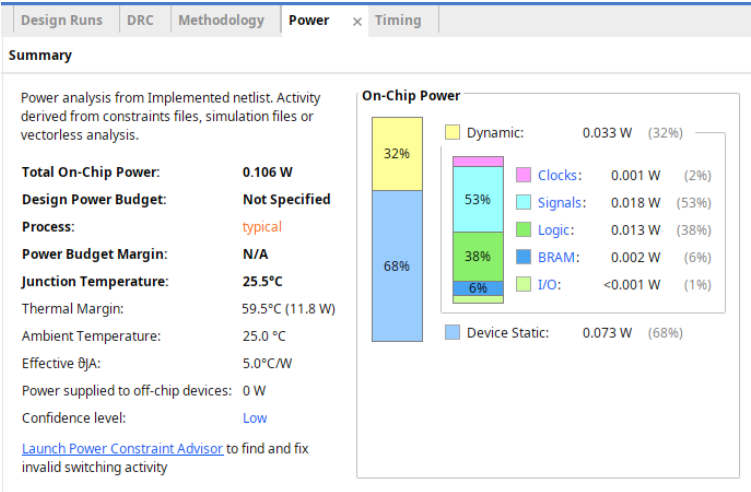


Figure 7: Power Summary

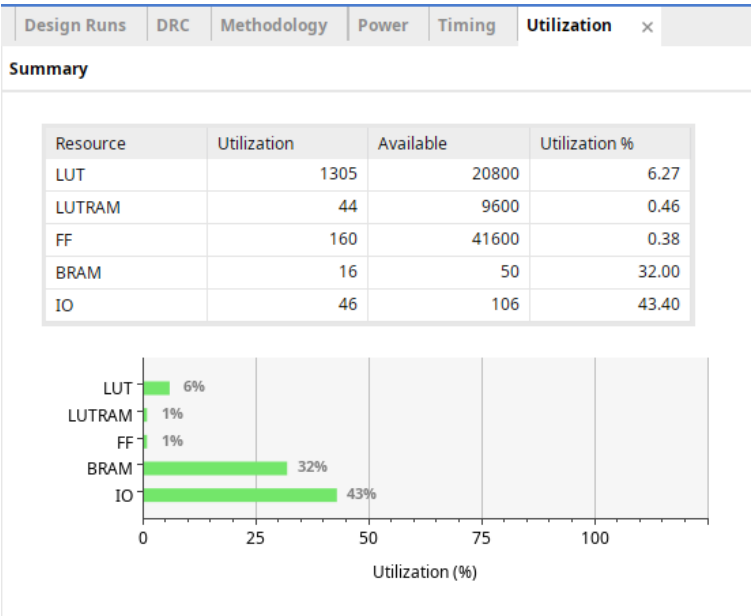


Figure 8: Utilization Summary

0.2 Testall Performance

0.2.1 Procedure

1. Load the testall mem file onto the multicycle otter. It should be noted that Diego’s testall program worked well such that the Basys3 board also gave LED and sseg indicators that the system was working.
2. Capture the same waveforms, cpu execution time, power, and area utilization.

0.2.2 Results

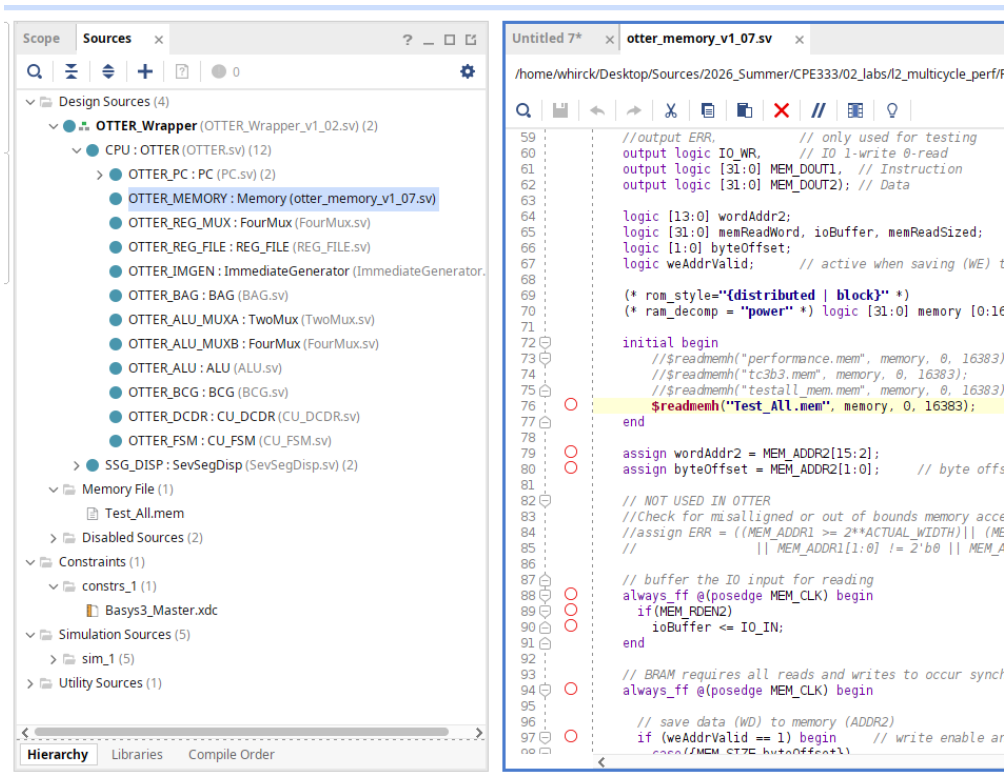


Figure 9: Uploading the testall memory

Timing

Clock Summary			
Name	Waveform	Period (ns)	Frequency (MHz)
sys_clk_pin	{0.000 5.000}	10.000	100.000

Figure 10: Clock Speed

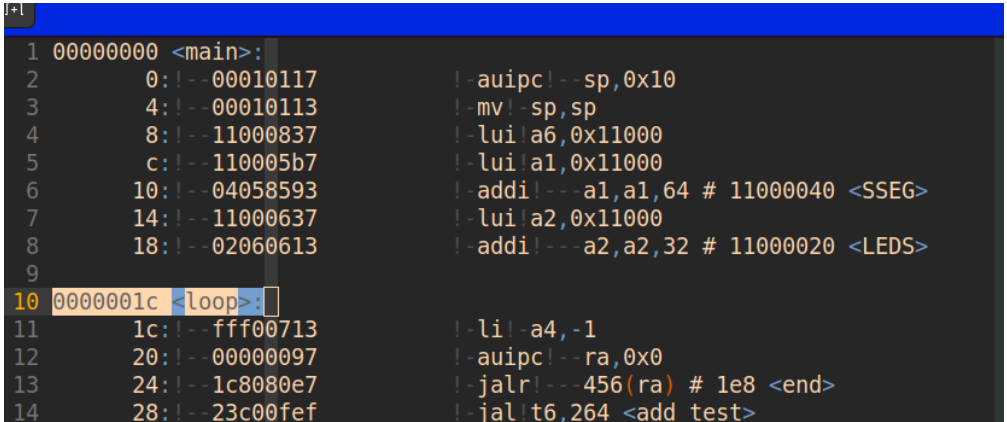


Figure 11: Target Instruction Address for Timing Intervals in `Test_All_Debug.txt`

The `loop` label was used for measurement because the testall program returns to this after running all tests.



Figure 12: Timestamps for loop label

$$\begin{aligned}
 t_{execution} &= t_{end} - t_{start} \\
 &= 834.2\mu s - 417.32\mu s \\
 &= 416.88\mu s
 \end{aligned}$$

$$\begin{aligned}
 n_{CLK_Cycles} &= \frac{t_{execution}}{T_{CLK}} \\
 &= \frac{416880ns}{10ns} \\
 &= 41,688
 \end{aligned}$$

Note that the number above is not the number of instructions executed.

Other Metrics

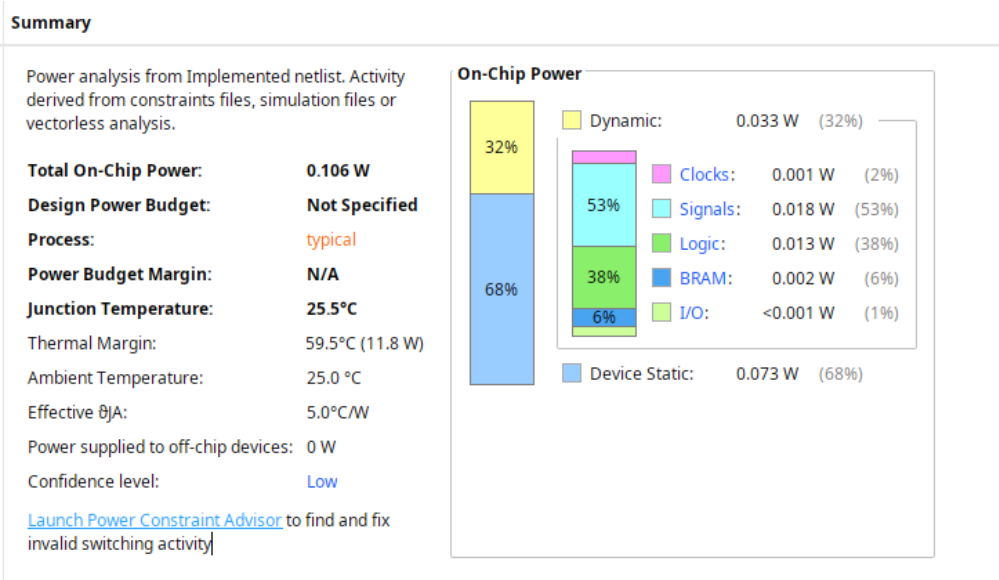


Figure 13: Power Summary of Testall on Basys3

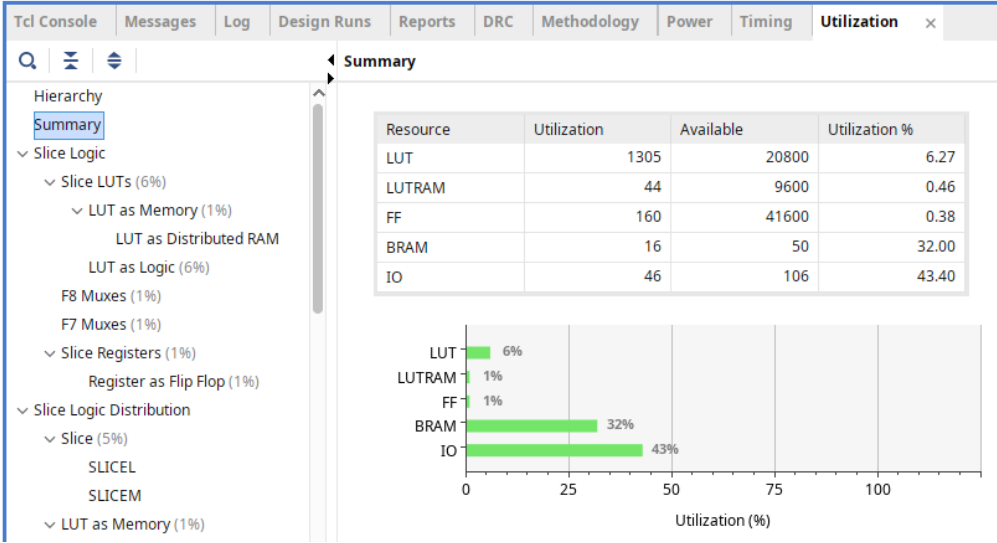


Figure 14: Utilization Summary of Testall

0.3 Notes

0.3.1 RARS

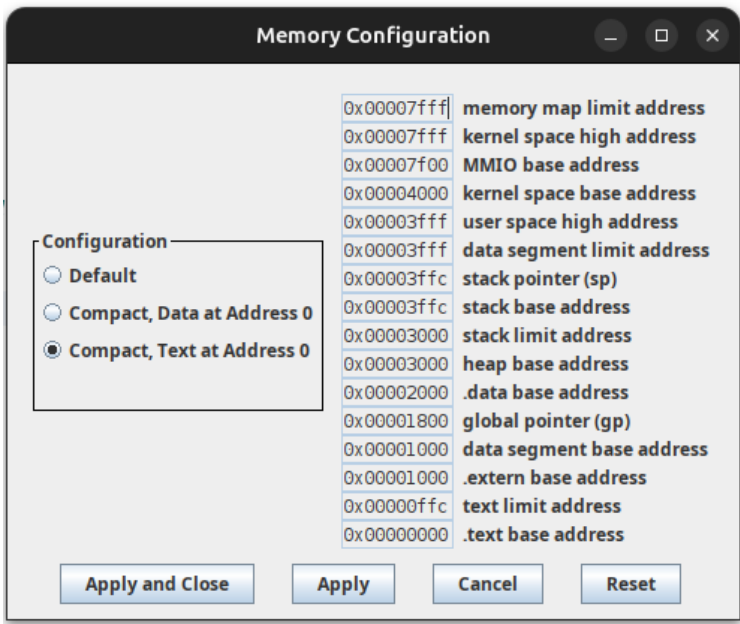


Figure 15: RARS Memory Config Settings

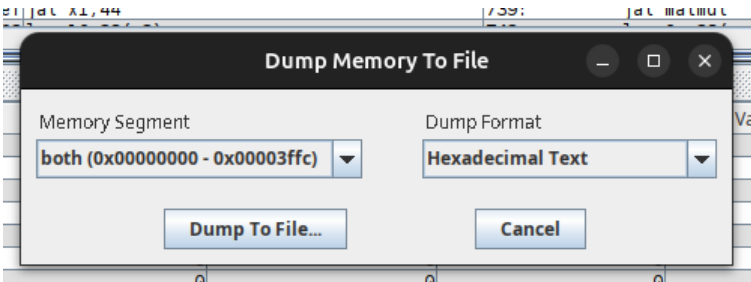


Figure 16: RARS Memory Dump Config

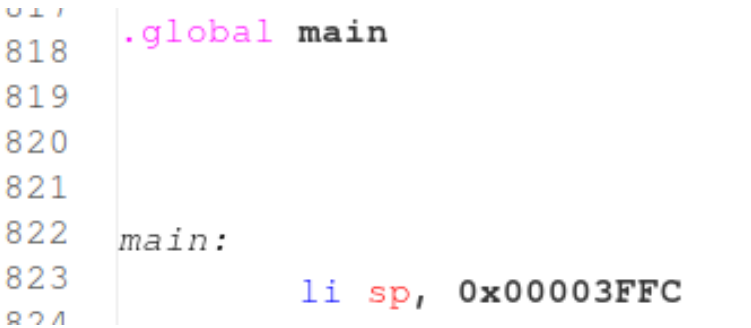


Figure 17: Stack Pointer set after Data Segment